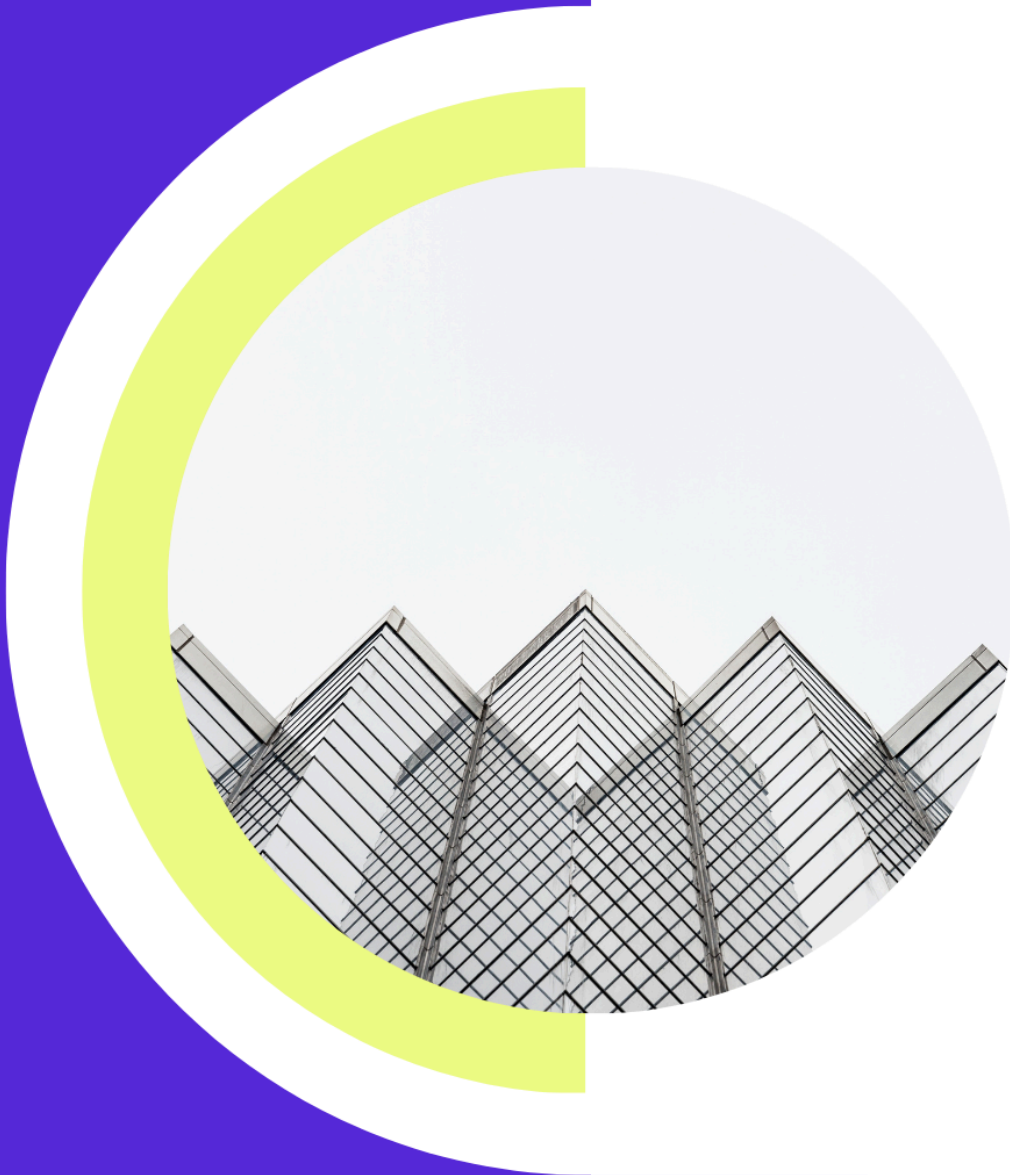




UNLOCKIT TRUST LAYER

Canton-powered Data Rail for Real Estate

Tiago Dias

Founder & CEO

917 800 029

tiago.dias@unlockit.io

Luís Marado

Co-Founder & CTO

931 767 206

luis.marado@unlockit.io

Real Estate Transaction Verification Network	3
Rationale and Context	3
Our Unique Value Proposition	3
Strengths	3
Ultimate Objectives	4
Why Canton	4
On-Chain vs Off-Chain	4
The Three Rails of the Trust Layer	5
<i>Verification Rail</i>	5
<i>Transactional Rail</i>	5
<i>Reputation Rail</i>	6
Canton Coin	6
Unique Property Registry	6
Connected Data Sources	6
Data Quality Standard (DQL)	7
<i>DQL Scale</i>	7
Trust Layer Model	7
Aggregation Data Layer	7
Verification and Authentication	8
Revenue Distribution Layer	8
<i>Distribution Factors</i>	8
<i>Integrity Controls</i>	8
Dashboard	8
Required Properties of the Shared Ledger	9
Application Layer	9
Iris — 360° Real Estate Transaction Engine	9
Clara — Smart Profiles and Smart Reviews	9
Other Modules	9
Vision: The First Fintech Layer on Top of a House	10

Real Estate Transaction Verification Network

Unlockit builds the baseline infrastructure that makes real estate verifiable, permissioned, and programmable. It sits before tokenization, not after it: the network standardizes proof so any transaction or financial product can run on audited facts without exposing sensitive deal data.

Unlockit's positioning is intentionally narrow and ownable. It operates as the verification rail for finance-ready real estate, enabling institutions to rely on standardized proof rather than bespoke diligence.

Rationale and Context

Real estate needs a better way to self-regulate. That starts with sovereign, high-quality data: immutable, privacy-preserving, and verifiably accurate. When information becomes independently verifiable, it stops being a compliance cost and becomes an ownable, priced asset — something participants can contribute to, stake reputation on, and earn from when it is used.

Institutions do not buy “tokens”. They buy proof: verifiable ownership, verifiable cash flow, verifiable compliance, and verifiable counterparties. When readiness becomes standard and repeatable, liquidity becomes a consequence, not a promise.

Our Unique Value Proposition

Unlockit is the trust data infrastructure for real estate: a decentralized Canton-based rail where profiles, reviews, verifications, and transactions become verifiable digital assets.

The network creates a shared ledger where **reference data** (what a property is), **transactional data** (what actually happened), and **verified participant profiles** (who did what) coexist as smart contracts with fine-grained privacy controls. This allows data to be reused across systems without centralizing ownership or exposing sensitive deal information.

Strengths

The Trust Layer is designed around a set of core strengths that directly address fragmentation, opacity, and inefficiency in real estate data exchange. Together, these capabilities enable verification, reuse, and monetization without sacrificing privacy or control.

- **Enhanced Transparency (verifiable, not public)** — provable facts with an auditable trail
- **Privacy by Design (selective visibility)** — sensitive fields are accessible only to required parties
- **Collaboration and Consistency** — data is validated once and reused; all changes remain traceable
- **Interoperability Backbone** — a single integration enables publishing and consumption across systems
- **Verified Identity and Reputation** — transaction-linked, attributable trust signals
- **Data Liquidity** — data is owned, priced, and paid for when used

Ultimate Objectives

The Trust Layer is guided by a set of long-term objectives that shape both its technical architecture and its participation model. These objectives define the outcomes the network is designed to unlock for the real estate ecosystem.

- Unlock real-estate information liquidity
- Enable access to aggregated, verifiable data with declared quality levels
- Guarantee stakeholder sovereignty over information
- Enforce equitable participation rules across the network

Why Canton

Real estate transactions require selective disclosure. Many blockchain systems assume transparency by default, which conflicts with regulated environments. Canton enables multi-party coordination with privacy and compliance built in. Institutions such as brokers, banks, notaries, and lawyers can share a minimal, verifiable state without exposing private deal data.

Canton is therefore not used as a public data store. It functions as a shared trust machine that anchors state, proof, and permissions while allowing payload data to remain private.

On-Chain vs Off-Chain

To remain institution-ready, Unlockit follows a strict architectural rule:

- On-chain (Canton) contains shared state, proofs, attestations, and permissions.
- Off-chain systems store full documents, and any other heavy data.

Registering the state in Canton serves four structural goals.

1. Creates portable reputation that is reusable across transactions rather than trapped within a single platform.

2. Provides fraud resistance through tamper-evident audit trails and immutable attribution.
3. Produces monetizable data quality by recording provenance — who attested what, when, and under which rule set.
4. Increases execution speed by reducing repeated KYC checks, document validations, and manual verifications.

Canton is not just shared storage. It is a coordination and accountability engine.

The Three Rails of the Trust Layer

The Trust Layer is structured into three coordinated rails. Each rail captures a different dimension of trust: verification of capabilities, execution of transactions, and accumulation of reputation. Together they transform isolated real estate activities into reusable, verifiable financial signals.

The rails are interoperable but distinct in purpose. Credentials establish who can act. Transactions record what happened. Profiles reflect how participants performed over time.

Verification Rail

Credentials are the backbone of reusable trust. They are not merely identity statements but structured claims that combine identity, capability, compliance scope, issuer authority, validity window, and revocation status.

Examples include professional licenses for agents and notaries, KYC/AML attestations, proof-of-funds confirmations issued by banks as claims rather than documents, and property-level checks such as title, lien, tax, or energy certificate verification.

On-chain, the system registers the signed credential metadata, issuer identity, validity period, and revocation state. Off-chain, supporting documents remain private. Credentials can be reused across transactions, dramatically reducing repeated compliance friction.

Transactional Rail

Real estate transactions have structured, verifiable workflows. On-chain, these connect to parties and roles, canonical property ids, workflow state transitions, obligations and conditions, human attestations, document commitments (hashes with access grants), settlement milestones, and the immutable audit timeline.

The workflow state machine covers stages such as offer, acceptance, due diligence, financing, and closing. Obligations function as compliance gates that must be satisfied before progression. Attestations convert manual validations into reusable proof.

Raw documents remain off-chain. The ledger records commitments and coordination signals rather than complete payload content.

Reputation Rail

Profiles anchor professional credibility to verified identity and transaction history. On-chain, the system registers credential links, transaction-linked feedback, derived performance signals, ranking computations based on verifiable inputs, multi-attester validations, and a persistent reputation timeline.

Only participants in a real transaction can issue transaction-linked reviews. Performance metrics such as time-to-close and dispute frequency are derived from transaction workflow history. Reputation evolves over time and cannot be reset after misconduct, ensuring durable accountability.

Canton Coin

Unlockit maintains a Canton Coin treasury as operating inventory for network participation and Global Synchronizer fees. Canton's burn-mint equilibrium model ties token economics to real usage. Unlockit treats CC as infrastructure fuel that supports long-term execution of the Trust Layer vision rather than speculative inventory.

Unique Property Registry

The Unique Property Registry is the foundation of the Trust Layer: a secure, permissioned ledger that maintains a unique record for each property. It creates a persistent property identity and an auditable lifecycle record built from authenticated inputs, provenance, and selective visibility.

Each property record includes:

- **Property identity** — canonical mapping across systems
- **Verifiable claims** — attributes, documents, milestones, validations
- **Provenance** — who asserted what, when, and under which permissions
- **Selective visibility** — confidentiality preserved while auditability remains

This is not a public database. It is a verification registry designed for real transactions where confidentiality, compliance, and institutional controls are required.

Connected Data Sources

The Property Registry aggregates information from multiple authoritative and accountable sources rather than relying on a single system of record. Each source contributes evidence under defined permissions and attribution rules.

- **Industry systems (MLS, CRM, back office)** — operational property and deal data linked to verified profiles
- **Open government data** — authoritative anchors for baseline facts
- **Collective intelligence via Iris** — accountable feedback from verified transaction participants
- **Transaction engine evidence** — milestones, validations, and workflow proof
- **Permissioned identity and document data** — accessible only to authorized parties for specific verification tasks

Data Quality Standard (DQL)

Every data item is assigned a **Data Quality Level (DQL)**, a standardized, auditable confidence grade so humans, institutions, and automated systems can immediately understand reliability.

Each claim carries a proof trail:

- **Provenance** — source, timestamp, and permissions
- **Evidence** — documents, references, certificates, and cryptographic anchors
- **Attestations** — who verified the claim and by which method
- **Freshness** — last verification date and renewal expectations
- **Dispute state** — challenged, unresolved, or resolved

DQL Scale

Data Quality Levels are expressed on a graduated scale that reflects increasing verification rigor and institutional readiness. Higher levels signal stronger assurance and broader usability.

- **DQL0** — Unverified
- **DQL1** — Anchored (tamper-evident provenance)
- **DQL2** — Attested (one authorized verifier or trusted system of record)
- **DQL3** — Corroborated (multiple independent confirmations)
- **DQL4** — Authority-grade (regulated issuer plus freshness)

Trust Layer Model

The Trust Layer operates as a modular, end-to-end system that transforms raw inputs into verifiable, monetizable proof. Each layer plays a distinct role while remaining interoperable with the others.

Aggregation Data Layer

Rather than requiring thousands of bilateral integrations, the aggregation layer provides a single shared rail. It unifies datasets, preserves lineage, supports DQL, and enables reuse without centralizing ownership.

Verification and Authentication

Verification is inseparable from identity. All inputs are tied to verified actors aligned with trusted identity providers, ensuring accountability from the outset.

Verification upgrades raw inputs into proof through role-based attestations, institutional validation, and due-process mechanisms backed by enforceable penalties.

Revenue Distribution Layer

The Trust Layer introduces deterministic monetization for verified data. When data is consumed by a workflow or purchased as a product, contributors are compensated automatically.

A fixed share of revenue is allocated to a Data Monetization Pool, which is distributed to eligible stakeholders based on their weighted contribution to the verified history.

Distribution Factors

Revenue allocation is calculated using a deterministic model that rewards accuracy, recency, and sustained contribution.

- **Time decay factor** — newer data carries more value
- **Transaction value weighting** — higher-value transactions have greater impact
- **Contribution consistency** — long-term high-quality contributors earn a larger share

Integrity Controls

To preserve trust and prevent manipulation, the distribution model is reinforced by integrity controls applied before and after compensation.

- Pre-compensation validation of inputs
- Retroactive penalization for proven inaccuracies
- Reversible and vesting-based rewards
- Dynamic reputation weighting

The outcome is a system where contributors retain sovereignty over their data, earn when it is used, and consumers pay for proof rather than manual verification.

Dashboard

Participation is managed through a unified dashboard that provides visibility into permissions, DQL levels, audit trails, controlled sharing, and earnings. This ensures transparency without sacrificing control.

Required Properties of the Shared Ledger

A shared ledger is a structural requirement rather than a design preference. The Trust Layer depends on properties that cannot be achieved through centralized databases or bilateral integrations.

- Tamper-evident shared audit trails
- Immutable attribution and lineage
- Programmable, deterministic enforcement
- No unilateral control over rules or history

Application Layer

The application layer exposes the Trust Layer's capabilities through user-facing products that generate reusable proof as a by-product of normal transaction workflows.

Iris — 360° Real Estate Transaction Engine

Iris converts real estate transactions into structured, verifiable workflows. Documents, validations, approvals, signatures, and evidence are captured around verified identities and property facts with selective visibility by design.

It integrates with existing CRMs and portals while adding a verification-grade workflow that produces durable proof.

Clara — Smart Profiles and Smart Reviews

Clara anchors professional credibility to verified identity, credentials, and transaction-linked evidence. It produces portable Smart Profiles and supports professional-to-professional and client-to-professional reviews.

Eligibility is proven through transaction context, with dispute resolution supported by verifiable evidence.

Other Modules

Beyond core workflows, the Trust Layer supports optional smart contract modules that can be enabled per transaction to automate specific outcomes.

- **Smart Rent** — automated rental agreements and enforcement

- **Smart Commission and Reserves** — conditional release of deposits and reserves
- **Smart Mortgage** — underwriting and covenant monitoring based on verified facts
- **Organizational Governance** — role-based approvals and compliance tracking
- **Payment and Settlement Integrations** — conditional fund release
- **Extended Instruments** — DeFi and TradFi products enabled by verifiable facts

Vision: The First Fintech Layer on Top of a House

Unlockit makes real estate finance-ready by default. When the market can prove who is involved, what the asset is, what happened to it, and what cash flows exist under defined rights and obligations, financial products no longer depend on fragmented evidence and manual interpretation.

The result is lower diligence friction, reduced fraud risk, and scalable financial services built on proof.